

Optimization and Evaluation of Machine Learning Classifiers on the Fashion-MNIST Dataset

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Abstract—This report presents an experimental evaluation of five machine learning classifiers applied to the Fashion-MNIST object recognition benchmark. Three feature representations were investigated: Raw intensities, Histogram of Oriented Gradients (HOG), and Principal Component Analysis (PCA). classifiers evaluated included k -Nearest Neighbors, Support Vector Machine, Logistic Regression, Multilayer Perceptron, and Convolutional Neural Networks. The implemented deep CNN model achieved the highest accuracy of 93.01% and a macro-F1 score of 93.0%, while k -NN with HOG achieved the best performance among all classical methods at 86.6%. The results presented in this study provide valuable insights into the trade-offs and dependencies between model complexity, feature representation, and performance for multi-class object recognition.

Index Terms—Machine Learning, Object Recognition

I. INTRODUCTION

This project explores different machine learning algorithms for object classification using the Fashion-MNIST dataset. The aim is to experimentally evaluate how feature engineering choices, classification techniques, and hyperparameter tuning can affect recognition performance. The study compares the implemented models under different preprocessing and tuning settings in order to understand their strengths, limitations, and ability to generalize to unseen data.

The report is organized as follows: Section II describes the dataset representation, preprocessing, and feature engineering. Section III presents the experimental setup, tuning procedure, and classification results for each model. Section IV discusses the comparative performance of the models. Section V concludes the report and outlines possible future improvements to address current limitations.

II. PREPROCESSING AND FEATURE ENGINEERING

The open-source Fashion-MNIST dataset was downloaded from the official Zalando Research repository [1]. This dataset consists of 70,000 grayscale images of fashion items, originally divided into 60,000 training images and 10,000 test images. Each image has a size of 28×28 pixels and belongs to one of ten classes.

The training set was further split into training and validation subsets to allow hyperparameter tuning of Neural Network (NN)-based models. Pixels were then normalized by dividing

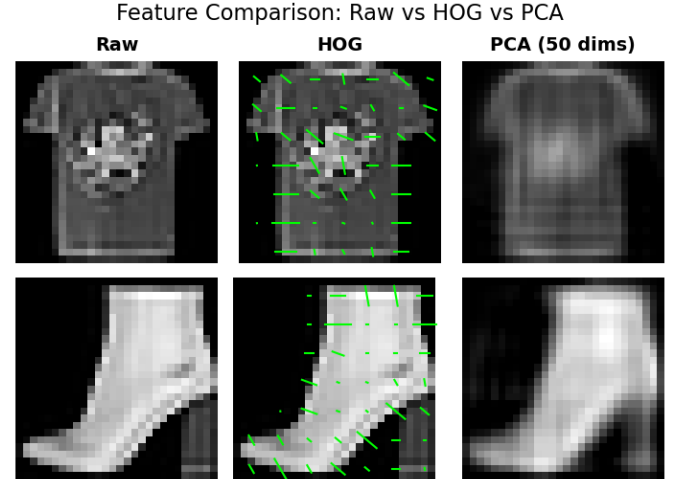


Fig. 1. Comparison of the feature representations used in this project: raw image, HOG edge-orientation representation, and PCA reconstruction.

their values by 255. In the raw feature representation, each image matrix was flattened into the vector:

$$\mathbf{x}_i = [x_{i1}, x_{i2}, \dots, x_{i784}]^T \in \mathbb{R}^{784}. \quad (1)$$

Three feature representations were then considered: raw pixel intensities, *Principal Component Analysis (PCA)* [2], and *Histogram of Oriented Gradients (HOG)* [3]. Raw pixels preserve the original grayscale intensity information with no change. PCA provides a compact representation by projecting the flattened image vectors onto the directions of highest variance, while HOG describes local edge and shape information through gradient orientation histograms. A visual comparison of the resulting representations is shown in Fig. 1.

For PCA, the image vectors were first plotted with the first two principal components to visualize the main directions of variation in the data, as shown in Fig. 2.

The number of principal components n was then selected to balance dimensionality reduction with meaningful feature representation. With $n = 50$, PCA captured 86.3% of the total variance, as shown in Fig. 3. This reduced the input dimension from 784 to 50, giving a compression factor of approximately 15.7 while preserving most of the data variation.

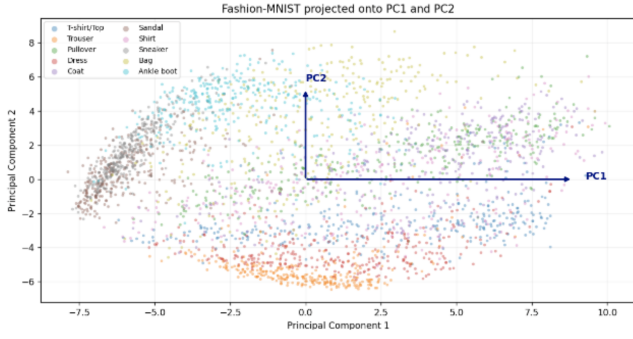


Fig. 2. Fashion-MNIST samples projected onto the first two principal components.

HOG features were also extracted to represent local shape and edge information instead of relying only on pixel intensities. In this method, we compute image gradients, groups them into orientation histograms, and concatenates them into one feature vector. With a cell size of 4×4 pixels and 9 orientation bins, each image produced a HOG feature vector of dimension 441. This representation is useful because fashion categories differ by outlines, edges, and local geometric structure rather than plain pixel values.

III. EXPERIMENTAL RESULTS

Experiments were conducted in Python 3.11 on an Intel Core i7-1255U processor and 8 GB RAM with no GPU acceleration. Most classifiers were implemented from scratch using NumPy and other libraries, while Logistic Regression and CNN were implemented using PyTorch.

All models were evaluated on the same test set of 10,000 images. The main reported metrics are the overall classification accuracy, which is the averaged percentage of correctly classified test samples, and macro-averaged F1 scores with equal per-class-weights were also reported.

A. Classical Machine Learning Models

1) *k*-Nearest Neighbors (*k*-NN): The *k*-Nearest Neighbors classifier is one of the simplest, non-parametric supervised

classification methods [4], as it does not assume a specific data distribution and does not require an explicit training phase. Instead, it memorizes the training data and predicts the label of a new sample based on the labels of its nearest neighbors. For this reason, *k*-NN is often described as a "lazy" learner.

In this implementation, distance between samples was measured using the Euclidean distance:

$$d(\mathbf{x}_i, \mathbf{x}_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2. \quad (2)$$

The predicted class is then selected based on the majority of the closest *k* training samples. To determine a best suitable value of *k*, the parameter was swept over $k \in \{1, 3, 5, 7, 9\}$. As shown in Fig. 4, the best result of the *k*-NN classifier on the raw feature representation was obtained at *k* = 5.

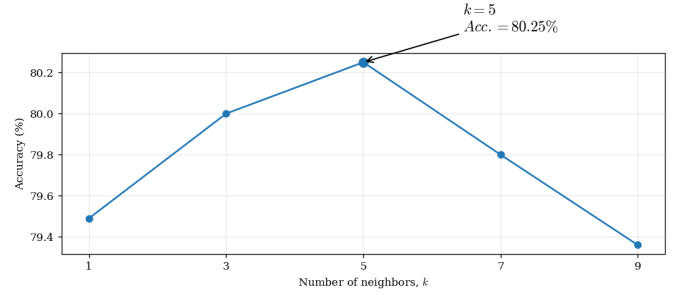


Fig. 4. *k*-NN accuracy for different values of *k* using raw pixel features.

Overall, *k*-NN showed strong performance despite its simplicity, achieving the highest accuracy of 86.6% on the HOG feature. Its main limitation, however, was its high prediction cost. Additionally, distance-based decisions do not necessarily correspond to actual semantic similarity between classes.

2) *Support Vector Machine (SVM)*: Support Vector Machine (SVM) is another supervised classification method that separates classes by finding a decision boundary in the feature space [5]. In the linear kernel case, this boundary is represented by the hyperplane:

$$\mathbf{w}^T \mathbf{x} + b = 0, \quad (3)$$

where $\mathbf{w}$ is the weight vector and b is the bias term. The goal is to choose the boundary that maximizes the margin between the classes, where the margin is determined by the closest samples to the hyperplane, known as *support vectors*.

SVM showed slightly lower accuracies on the raw and PCA feature representations, achieving 82.3% and 77.0% respectively. However, unlike with *k*-NNs, its performance decreased more when used on the HOG feature, which suggests that the selected SVM formulation is more effective on pixel intensities representations. This is likely due to the main limitation of this method, which is that a linear boundary cannot fully capture the nonlinear differences between the visually similar fashion classes. [5]

3) *Logistic Regression*: In Logistic Regression, the model learns a separate score for each of the ten Fashion-MNIST classes on each input feature vector:

$$z_k = \mathbf{w}_k^T \mathbf{x} + b_k, \quad k = 0, 1, \dots, 9, \quad (4)$$

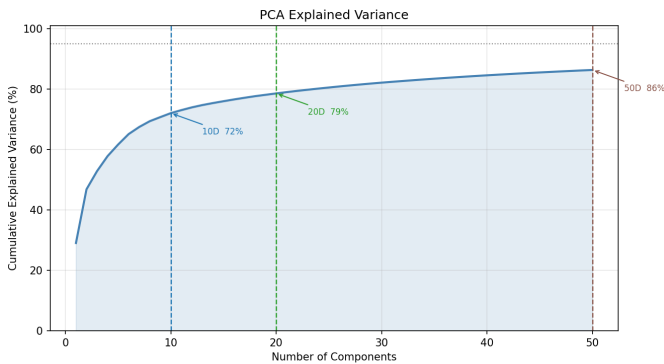


Fig. 3. Cumulative explained variance of PCA components used to select a reduced feature dimension.

where $\mathbf{w}_k$ is the learned weight vector for class k , b_k is the bias term, and z_k is the class score.

These scores are then converted into probabilities using the softmax function:

$$P(y = k|\mathbf{x}) = \frac{e^{z_k}}{\sum_{j=0}^9 e^{z_j}}. \quad (5)$$

The predicted label is then selected as the class with the highest probability amongst all.

The Logistic Regression implementation was trained using the Adam optimization [6]. The results showed similar performance for raw and HOG features, with accuracies of 84.6% and 84.5%, respectively. This indicates that the model was able to benefit from both very different representations effectively.

Table I summarizes the performance of all three classical machine learning models across the tested feature representations. Overall, k -NN achieved the best result (see Fig 5, especially when combined with HOG features).

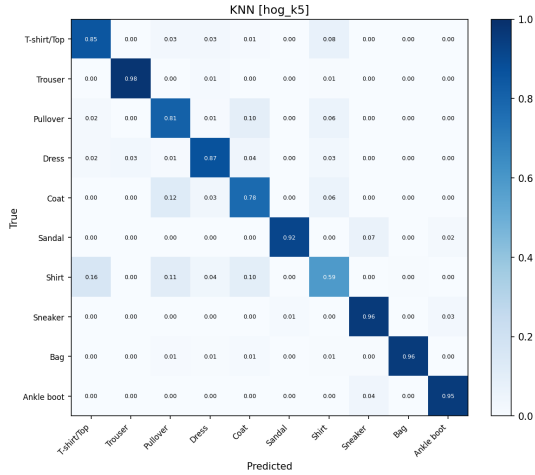


Fig. 5. Confusion matrix of the best 5-NN configuration using HOG features.

TABLE I
PERFORMANCE RESULTS OF CLASSICAL MACHINE LEARNING MODELS

Classification Model	Raw		PCA		HOG	
	Acc. (%)	Macro-F1	Acc. (%)	Macro-F1	Acc. (%)	Macro-F1
k -NN ($k = 5$)	85.1	85.0	85.5	85.4	86.6	86.5
SVM	82.3	81.8	77.0	74.5	71.9	69.9
Logistic Regression	84.6	84.5	82.3	82.2	84.5	84.4

B. Neural Network Models

Neural network (NN) models were also implemented to compare classical feature-based approaches with more complex models that can learn nonlinear representations. Two NN-based models were evaluated: a Multilayer Perceptron (MLP) model, and a Convolutional Neural Network (CNN) model.

1) *Multilayer Perceptron (MLP)*: MLP is a feedforward neural network that extends linear classification by adding hidden layers and nonlinear activation functions [7]. Unlike the implemented logistic regression model that learned a linear

decision boundary, the MLP model can adapt to nonlinear relationships between input features and output classes.

In this implementation, MLP was built using fully connected layers with a ReLU activation functions. Dropout and L2 regularization were integrated to prevent overfitting, and an early stopping was used to stop training when the validation performance stopped improving for five consecutive epochs. The model was trained on both raw and PCA representations to compare the effect of dimensionality reduction on its performance.

MLP achieved strong performance compared to all classical models, reaching 90.0% accuracy and macro-F1 score using raw features only. The PCA representation gave a slightly lower accuracy of 88.5%, indicating that the reduced feature vector might have removed some underlying information that could have improved classification performance.

2) *Convolutional Neural Network (CNN)*: Finally, a Convolutional Neural Network (CNN) was implemented. The CNN was expected to outperform all previous models as it is the only one that takes advantage of the spatial information of 2D images. It applies learnable convolutional filters over local image regions to extract feature maps, where earlier layers capture simple, low-level patterns such as edges, and deeper layers then combine them into more meaningful visual features.

For this experiment, three CNN architectures were tested: small, medium, and deep, as summarized in Table III. All architectures were evaluated on raw image matrices only.

TABLE II
THREE IMPLEMENTED CNN ARCHITECTURES

	Small	Medium	Deep
Architecture	1 conv. layer, 32 maps	2 conv. layers, 32-64 maps	3 conv. layers, 32-64-128 maps

During tuning, the deep CNN model showed signs of overfitting (see Fig. 6), where the training loss kept decreasing while the validation loss began to rapidly increase. To solve this, the dropout rate was increased from 0.2 to 0.4, meaning that 40% of neurons were randomly disabled during training. This reduces co-adaptation between neurons and can therefore improve generalization [8]. The final results show all CNN models achieving above 90% accuracies, with the highest score of 93.01% by the deep CNN model.

A summary of all results of NN-based models is presented in Table III

IV. DISCUSSION OF RESULTS

Among classical methods, k -NN achieved the strongest performance, reaching 86.6% accuracy with HOG features. This result highlights the effectiveness of shape-based representations for fashion item recognition, as HOG captures local edge orientations that correspond well to the structural differences between clothing categories. Logistic Regression performed consistently across raw and HOG features at around 84.5%, indicating that the linear decision boundary it learns is somewhat feature-agnostic for this dataset.

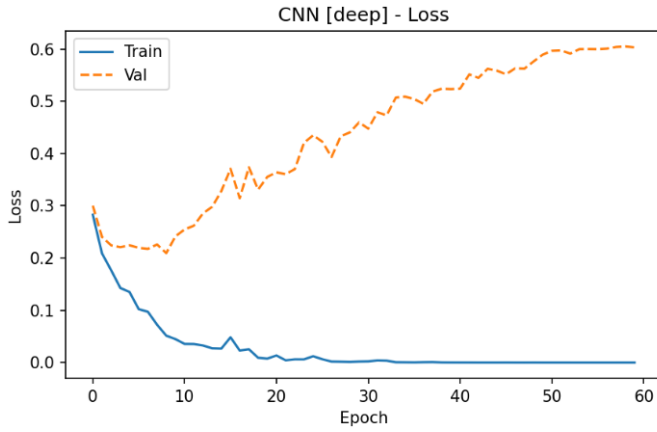


Fig. 6. Train loss vs. validation loss, showing overfitting

TABLE III
PERFORMANCE RESULTS OF NEURAL NETWORK MODELS

Multilayer Perceptron (MLP)		
Feature	Acc. (%)	Macro-F1 (%)
Raw	90.00	90.00
PCA	88.50	88.40
Convolutional Neural Network (CNN), Raw Images		
Architecture	Acc. (%)	Macro-F1 (%)
Small	91.17	91.10
Medium	92.93	92.90
Deep	93.01	93.00

It was observed that PCA generally reduced accuracy for most classifiers compared to raw pixel features, except for k -NNs, where it provided a noticeable improvement in performance. This suggests that Fashion-MNIST is already relatively clean and low-dimensional, so most of the discarded low-variance components may have still contained useful discriminative details. For simpler classifiers that rely directly on pixel-level differences, removing these details have reduced classification performance.

However, PCA was still useful from a computational point of view. By reducing the input dimension from 784 to 50, it significantly lowered the feature size, which improved efficiency for models such as k -NN and SVM. Therefore, PCA provided a trade-off: a small loss in accuracy for most models, but a clear reduction in computational cost.

Neural network models outperformed all classical methods, as expected given their ability to model nonlinear decision boundaries. The MLP model achieved 90.0% accuracy on raw features, substantially exceeding the best classical result of 85.1% (for 5-NN). The CNN further improved this to 93.01% with the deep architecture, and no feature engineering, benefiting from its ability to exploit the spatial structure of 2D images through learned convolutional filters. The gap between CNN and all other methods underscores the importance of pre-

serving spatial information when working with 2-dimensional data.

Across all models, the Shirt and T-shirt/Top classes consistently produced the most misclassifications, as evidenced by their lower per-class F1 scores. This is expected given their visual similarity with multiple other classes. The CNN confusion matrix showed the clearest separation between these classes, further demonstrating the advantage of spatially-aware feature learning.

V. CONCLUSION AND FUTURE WORK

This project implemented and evaluated five machine learning classifiers on the Fashion-MNIST dataset using different feature representations, including raw pixels, PCA-reduced features, and HOG features. The results showed that neural-network-based methods achieved the strongest overall performance, with the deep CNN obtaining the best result of 93.01% accuracy. This confirms the advantage of convolutional models for image classification, since they can preserve and learn from the spatial structure of the input images.

Among classical methods, k -NN achieved the best performance when combined with HOG features. This suggests that edge and shape-based representations were useful for distance-based classification. Logistic Regression also provided a strong linear baseline, while SVM performance was more sensitive to the selected feature representation. PCA reduced the computational cost by compressing the input dimension, but it generally caused a small accuracy drop for most classifiers.

Future work could focus on improving the CNN architecture through additional hyperparameter tuning, and exploring non-linear kernels for SVMs.

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